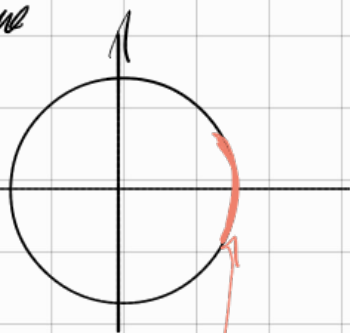


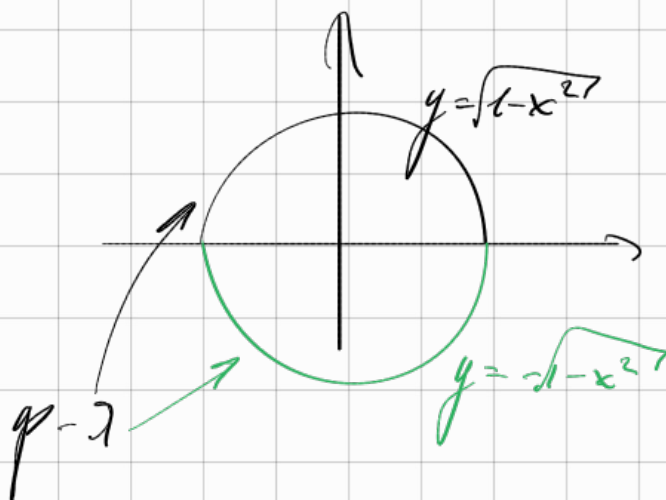
Реальные функции
 $F(x,y) = 0$ - уравнение

$$x^2 + y^2 = 1$$

не
 функция



не ф-я



Теорема о неявной ф-ии

Если $F(x,y)$ нем $\in U_0(x_0, y_0)$ $F(x_0, y_0) = 0$ и F'_y нем $\in U_0(x_0, y_0) \cup F'_y(x_0, y_0) \neq 0$, тогда $\exists U_1(x_0)$ и $U_2(y_0)$ $\forall x \in U_1(x_0) \exists y \in U_2(y_0) \rightarrow F(x,y) = 0$
 Задача $y = f(x)$

Следствие

Если в точке $F'_x \neq 0$ на $U_0(x_0, y_0)$ и нем на $U_0(x_0, y_0)$, то f нем $\in x_0$, тогда

$$f'(x_0) = -\frac{F'_x(x_0, y_0)}{F'_y(x_0, y_0)}$$

Если в точке F'_x и F'_y нем $\in U_0(x_0, y_0)$, то f в нек. окр. имеет нем. производную:

$$f'(x) = -\frac{F'_x(x, f(x))}{F'_y(x, f(x))}$$

$$T1 \quad x^2 = y^2$$

а) Сколько функций $y: \mathbb{R} \rightarrow \mathbb{R}$ удовлетворяют?



далее решаем

б) Сколько непрерывных ф-ий $y: \mathbb{R} \rightarrow \mathbb{R}$ удовлетворяют?

4



- 6) проверить нем. ф-ии $y: \mathbb{R} \rightarrow \mathbb{R}$ чтобы был $y(1)=1$? $\checkmark$
- 2) проверить нем. ф-ии $y: [1;2] \rightarrow \mathbb{R}$ чтобы был $y(1)=1$? $\checkmark$

Теорема

$F(x,y): U_0(x^{(0)}, y^{(0)}) \rightarrow \mathbb{R}$ $x^{(0)} \in \mathbb{R}^n$
 $y^{(0)} \in \mathbb{R}$
 Если $F(x,y)$ нем. группа $U_0(x^{(0)}, y^{(0)})$, $F(x^{(0)}, y^{(0)}) = 0$
 и $F'_y(x^{(0)}, y^{(0)}) \neq 0$, то $\exists U_1(x^{(0)}) \exists U_2(y^{(0)}) \forall x \in U_1(x^{(0)}) \exists (1) y \in U_2(y^{(0)}) \rightarrow$
 $\rightarrow F(x,y) = 0$ т.е. на $U_1(x^{(0)})$ группа $y = f(x)$
 Тогда можно найти группу в $U_1(x^{(0)})$ и $f'_x(x) = - \frac{F'_x(x; f(x))}{F'_y(x; f(x))}$

§ 3 № 61(а) $F(x,y,u) = 0$

$$x^2 - 2y^2 + 3u^2 - yu + y = 0$$

$$a(1; 1; \frac{1}{3}) \quad u(x,y)$$

$$F'_u = 6u - y$$

$$F'_u(1; 1; \frac{1}{3}) = 1 \neq 0$$

$$F'_x = 2x \quad F'_y = -4y - u + 1$$

$$F'_x(1; 1; \frac{1}{3}) = 2 \quad F'_y(1; 1; \frac{1}{3}) = -4 - \frac{1}{3} + 1 = -\frac{10}{3}$$

$$u'_x = - \frac{F'_x}{F'_u} = - \frac{2}{1} = -2$$

$$u'_y = - \frac{F'_y}{F'_u} = - \frac{-\frac{10}{3}}{1} = \frac{10}{3}$$

§ 3 № 64(а)

$$u^3 - xu + y = 0$$

$$a(3; -2; 2)$$

$$u'_x = - \frac{F'_x}{F'_u} = \frac{u}{3u^2 - x} = \frac{2}{12 - 3} = \frac{2}{9}$$

$$u'_y = - \frac{F'_y}{F'_u} = \frac{1}{3u^2 - x} = \frac{1}{9} \quad du = \frac{2}{9} dx - \frac{1}{9} dy$$

Система уравнений

$$F_1(x_1, \dots, x_n, y_1, \dots, y_m) = 0$$

...

$$F_m(x_1, \dots, x_n, y_1, \dots, y_m) = 0$$

$$\vec{x} = (x_1, \dots, x_n)^T$$

$$\vec{y} = (y_1, \dots, y_m)^T$$

$$\vec{F} = (F_1, \dots, F_m)^T$$

$$\vec{F}(\vec{x}; \vec{y}) = \vec{0}$$

$$\frac{\partial \vec{F}}{\partial \vec{y}} = \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \dots & \frac{\partial F_1}{\partial y_m} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial y_1} & \dots & \frac{\partial F_m}{\partial y_m} \end{pmatrix}$$

$$\frac{\partial \vec{F}}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \dots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}$$

Теорема

$\exists \vec{F}(\vec{x}; \vec{y})$ near gap of $U_{\delta_0}(\vec{x}^{(0)}; \vec{y}^{(0)})$, where $\vec{F}(\vec{x}^{(0)}; \vec{y}^{(0)}) = \vec{0}$ и

$$\det \frac{\partial \vec{F}}{\partial \vec{y}}(\vec{x}^{(0)}; \vec{y}^{(0)}) \neq 0$$

implies $\exists U_{\delta_1}(\vec{x}^{(0)}) \cup \exists U_{\delta_2}(\vec{y}^{(0)}): \forall \vec{x} \in U_{\delta_1}(\vec{x}^{(0)}) \exists (1) \vec{y} \in U_{\delta_2}(\vec{y}^{(0)}) \rightarrow \vec{F}(\vec{x}; \vec{y}) = \vec{0}$

i.e. on $U_{\delta_1}(\vec{x}^{(0)})$: $\vec{y} = \vec{f}(\vec{x})$, where $\vec{f}$ -near gap $\frac{\partial \vec{f}}{\partial \vec{x}} = - \left[\frac{\partial \vec{F}}{\partial \vec{y}} \right]^{-1} \left[\frac{\partial \vec{F}}{\partial \vec{x}} \right]$

§3 №25

(1;2) $u(x,y)$ $v(x,y)$

$$x e^{u+v} + 2uv = 1$$

$$y e^{u-v} - \frac{u}{1+v} = 2x$$

$$u(1;2) = v(1;2) = 0$$

$$F_1(x, y, u, v) = x e^{u+v} + 2uv - 1 = 0$$

$$F_2(x, y, u, v) = y e^{u-v} - \frac{u}{1+v} - 2x = 0$$

$$\frac{\partial \vec{F}}{\partial \vec{y}} = \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \det = -3$$

$$\frac{\partial F_1}{\partial u} = x e^{u+v} + 2v = 1 \quad \frac{\partial F_1}{\partial v} = x e^{u+v} + 2u = 1$$

$$\frac{\partial F_2}{\partial u} = y e^{u-v} - \frac{1}{1+v} = 1 \quad \frac{\partial F_2}{\partial v} = -y e^{u-v} \frac{u}{(1+v)^2} = -2$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\Delta} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \left(\frac{\partial \vec{F}}{\partial \vec{y}} \right) = \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} \frac{1}{-3} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\left(\frac{\partial \vec{F}}{\partial \vec{x}} \right) = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

$$\frac{\partial F_1}{\partial x} = e^{u+v} = 1 \quad \frac{\partial F_1}{\partial y} = 0 \quad \frac{\partial F_2}{\partial x} = -2 \quad \frac{\partial F_2}{\partial y} = e^{u-v} = 1$$

$$\vec{f} = \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u(x,y) \\ v(x,y) \end{pmatrix}$$

$$\frac{\partial \vec{f}}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} 0 & 1 \\ 3 & -1 \end{pmatrix}$$

$$\frac{\partial u}{\partial x} = 0 \quad \frac{\partial u}{\partial y} = -\frac{1}{3} \quad \frac{\partial v}{\partial x} = -1 \quad \frac{\partial v}{\partial y} = \frac{1}{3}$$

12.09

$$F(x,y,u) = 0$$

$$u'_x = - \frac{F'_x(x,y,u)}{F'_u(x,y,u)}$$

$$u'_y = - \frac{F'_y(x,y,u)}{F'_u(x,y,u)}$$

$$u''_{xx} = (u'_x)'_x = \left(\frac{F'_x(x,y,u(x,y))}{F'_u(x,y,u(x,y))} \right)'_x =$$

$$= - \frac{F''_{xx}(x,y,u(x,y)) + F''_{xu}(x,y,u(x,y)) u'_x(x,y) F'_u(x,y,u(x,y)) - F''_{ux}(x,y,u(x,y)) + (F'_u(x,y,u(x,y)))^2}{(F'_u(x,y,u(x,y)))^2}$$

$$+ \frac{F''_{uu}(x,y,u(x,y)) u'_x(x,y) F'_x(x,y,u(x,y))}{(F'_u(x,y,u(x,y)))^2}$$

§4 №43(5)

$$(x,y,u) = (1, -1, -1)$$

Найти $J^2 u$

$$u(1, -1) = -1$$

$$F(x, y, u) = u^3 + 2y^2u + xy = 0$$

$$F'_x = y \quad F'_y = 2u + x \quad F'_u = 3u^2 + 2y$$

$$F'_x|_{(1, -1, -1)} = -1$$

$$F'_y|_{(1, -1, -1)} = -1$$

$$F'_u|_{(1, -1, -1)} = 1$$

$$u'_x = -\frac{F'_x}{F'_u} = 1$$

$$u'_y = -\frac{F'_y}{F'_u} = 1$$

$$du|_{(1, -1)} = dx + dy$$

$$u''_{xx} = \frac{y}{(3u^2(x, y) + 2y)^2} (6u(x, y)u'_x(x, y))$$

$$u''_{xx}|_{(1, -1)} = 6$$

$$u''_{xy} = (u'_x)'_y = \frac{-(3u^2(x, y) + 2y) - 6u(x, y)u'_y(x, y) + 2y}{(3u^2(x, y) + 2y)^2} \Big|_{(1, -1)} = 3$$

$$u''_{yy} = \dots 2$$

$$J_u^2 \Big|_{(1, -1)} = 6dx^2 + 6dxdy + 2dy^2$$

§4 146(1)

$$f(x+u, y+u) = 0 \quad f \in C^2, u \in C^2$$

$$F(x, y, u) = f(x+u, y+u) = 0$$

$$f = f(a, b) \quad \frac{\partial f}{\partial a} = f'_1 \quad \frac{\partial f}{\partial b} = f'_2$$

$$F'_x = f'_1 \quad F'_y = f'_2 \quad F'_u = f'_1 + f'_2$$

$$u'_x = -\frac{F'_x}{F'_u} = -\frac{f'_1}{f'_1 + f'_2} \quad u'_y = -\frac{f'_2}{f'_1 + f'_2}$$

$$u'_x = -\frac{f'_1(x+u(x, y), y+u(x, y))}{f'_1(x+u(x, y), y+u(x, y)) + f'_2(x+u(x, y), y+u(x, y))}$$

$$du = -\frac{f'_1}{f'_1 + f'_2} dx - \frac{f'_2}{f'_1 + f'_2} dy$$

$$u''_{xx} = \frac{(f''_{11} [1+u'_x] + f''_{12} u'_x)(f'_1 + f'_2) - (f''_{11} [1+u'_x] + f''_{12} u'_x + f''_{21} [1+u'_x] + f''_{22} u'_x) f'_1}{(f'_1 + f'_2)^2}$$

$$u_{jj}'' = \frac{(f_{11}'' u_j' + f_{12}'' [1 + u_j']) (f_1' + f_2') - (f_{11}'' u_j' + f_{12}'' [1 + u_j'] + f_{21}'' u_j' + f_{22}'' [1 + u_j']) f_1'}{(f_1' + f_2')^2}$$

$$u_{jj}'' = \frac{(f_{21}'' u_j' + f_{22}'' [1 + u_j']) (f_1' + f_2') - (f_{11}'' u_j' + f_{12}'' [1 + u_j'] + f_{21}'' u_j' + f_{22}'' [1 + u_j']) f_2'}{(f_1' + f_2')^2}$$

Дифференцируемые отображения

Оп. $\exists$ ф. f нек. участка, отображающего каждую $x \in G \subset \mathbb{R}^n$ в элемент $y \in \mathbb{R}^m$ тогда f - отображение, если $f: G \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$f \Leftrightarrow \begin{cases} y_1 = f_1(x_1, \dots, x_n) \\ \vdots \\ y_m = f_m(x_1, \dots, x_n) \end{cases}$$

$f_i(x_1, \dots, x_n)$ координатные функции

Оп. (Диф) $f: U_\delta(x_0) \subset \mathbb{R}^n$
 $x_0 = (x_0^1; \dots; x_0^n)$

Тогда, если f диф. в т. x_0 , то $f(x) - f(x_0) = A \cdot \Delta x + o(|\Delta x|)$ $\Delta x \rightarrow 0$ $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

A - матрица $m \times n$

$$\Delta x = \begin{pmatrix} x^1 - x_0^1 \\ \vdots \\ x^n - x_0^n \end{pmatrix}$$

$$|\Delta x| = \sqrt{\sum_{i=1}^n (\Delta x^i)^2}$$

Замечание Если f диф. в x_0 $A = \begin{pmatrix} \frac{\partial f_1}{\partial x^1} & \dots & \frac{\partial f_1}{\partial x^n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x^1} & \dots & \frac{\partial f_m}{\partial x^n} \end{pmatrix}$ - матрица Якоби

Если $m=n$, то A - квадр.
 $\det A$ - якобиан

Теорема $\exists f: U_\delta(x_0) \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$

J - матрица Якоби

$\exists f$ - нек. диф. в этой окрестности, $\det J|_{x_0} \neq 0$

Тогда $\exists U_\varepsilon(y_0)$, $y_0 = f(x_0)$ отображение обратное отображению, которое не диф. в x_0

T3 $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$u = e^x \cos y$$

$$v = e^x \sin y$$

$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}$$

$$\det J = e^{2x} \cos^2 y + e^{2x} \sin^2 y = e^{2x} = 0$$

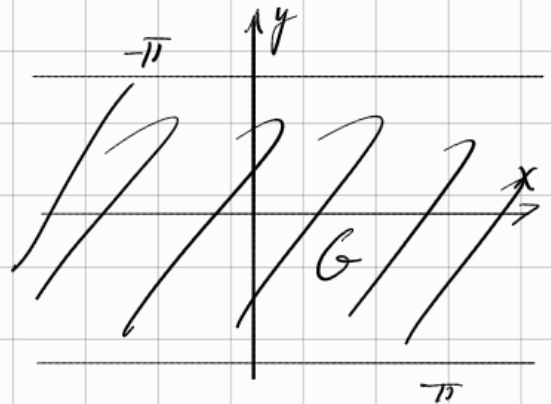
$$\begin{pmatrix} u \\ v \end{pmatrix}(x; y) = \begin{pmatrix} u \\ v \end{pmatrix}(x; y + 2\pi)$$

не является гомеоморфизмом на $\mathbb{R}^2$

на-бо значений единичным $\mathbb{R}^2 \setminus (0; 0)$

$$\text{Обл } G = \mathbb{R}^2 \setminus (0; 0)$$

$$\mathbb{R}^2 \setminus (0; 0) \xrightarrow[\text{гомеом}]{\text{бл}} (-\infty, +\infty) \times [-\pi; \pi]$$



T4

$$F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$x = r \cos \varphi$$

$$y = r \sin \varphi \quad r \geq 0$$

$$\Leftrightarrow \varphi = \varphi(x; y)$$

$$r = r(x; y)$$

$$\begin{matrix} r'_x & r'_y \\ \varphi'_x & \varphi'_y \end{matrix}$$

$$r = \sqrt{x^2 + y^2}$$

$$\tan \varphi = \frac{y}{x}$$

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2}} = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\tan \varphi = \frac{y}{x} \quad \frac{1}{\cos^2 \varphi} \cdot \varphi' = -\frac{y}{x^2}$$

$$\varphi' = -\frac{y \cos^2 \varphi}{x^2} = -\frac{y}{r^2}$$

$$\frac{1}{\cos^2 \varphi} \cdot \varphi' = \frac{1}{x}$$

$$\varphi' = \frac{\cos^2 \varphi}{x} = \frac{x}{r^2}$$

$$J = \begin{pmatrix} r'_x & r'_y \\ \varphi'_x & \varphi'_y \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}$$

Другой способ

$$y = f(x) \quad x = g(y) \quad J_f = J_g^{-1}$$

$$x = x(r, \varphi)$$

$$y = y(r, \varphi)$$

$$x_r = \cos \varphi$$

$$x_\varphi = -r \sin \varphi$$

$$y_r = \sin \varphi$$

$$y_\varphi = r \cos \varphi$$

$$J_{(r, \varphi) \rightarrow (x, y)} = \begin{pmatrix} x_r & x_\varphi \\ y_r & y_\varphi \end{pmatrix} = \begin{pmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{pmatrix}$$

$$\det J_{(r, \varphi) \rightarrow (x, y)} = r$$

$$J_{(x, y) \rightarrow (r, \varphi)} = J_{(r, \varphi) \rightarrow (x, y)}^{-1} = \frac{1}{\det} \begin{pmatrix} r \cos \varphi & r \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix} = \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\frac{\sin \varphi}{r} & \frac{\cos \varphi}{r} \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}$$

19.09

§ 3 / 86

$$x \frac{\partial \varphi}{\partial y} - y \frac{\partial \varphi}{\partial x} = 0$$

$$x = r \cos \varphi$$

$$y = r \sin \varphi$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial \varphi}{\partial r} = -\frac{y}{r^2}$$

$$\frac{\partial \varphi}{\partial y} = \frac{x}{r^2}$$

$$u = u(x, y) = \tilde{u}(r, \varphi) = \tilde{u}(r(x, y); \varphi(x, y))$$

$$\frac{\partial u}{\partial x} = \frac{\partial \tilde{u}}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial \tilde{u}}{\partial \varphi} \frac{\partial \varphi}{\partial x} = \frac{x}{r} \frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2} \frac{\partial \tilde{u}}{\partial \varphi}$$

$$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial \tilde{u}}{\partial \varphi} \frac{\partial \varphi}{\partial y} = \frac{y}{r} \frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2} \frac{\partial \tilde{u}}{\partial \varphi}$$

Polynomdivision

$$x \left(\frac{y}{r} \frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2} \frac{\partial \tilde{u}}{\partial \varphi} \right) - y \left(\frac{x}{r} \frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2} \frac{\partial \tilde{u}}{\partial \varphi} \right) = \cancel{\frac{xy}{r} \frac{\partial \tilde{u}}{\partial r}} + \frac{x^2}{r^2} \frac{\partial \tilde{u}}{\partial \varphi} - \cancel{\frac{xy}{r} \frac{\partial \tilde{u}}{\partial r}} + \frac{y^2}{r^2} \frac{\partial \tilde{u}}{\partial \varphi} =$$

$$= \frac{\partial \tilde{u}}{\partial \varphi} = 0$$

$$\tilde{u} = f(r) \quad f \in C^1_{(\text{nonneg.})}$$

$$u = f(\sqrt{x^2 + y^2}), \quad f \in C$$

§ 3 № 30

Телеграфное уравнение $(y-z)\frac{\partial z}{\partial x} + (y+z)\frac{\partial z}{\partial y} = 0$ найти x за y и z ,

a $u = y - z$
 $v = y + z$ за u, v

1) $y = \frac{v+u}{2}$
 $z = \frac{v-u}{2}$

$dy = \frac{1}{2}du + \frac{1}{2}dv$

$dz = -\frac{1}{2}du + \frac{1}{2}dv$

2) $x = x(u, v)$ - найти

$dx = \frac{\partial x}{\partial u}du + \frac{\partial x}{\partial v}dv$

3) $z = z(x, y)$ - найти

$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

4) 2) → 3)

$dz = \frac{\partial z}{\partial x}(\frac{\partial x}{\partial u}du + \frac{\partial x}{\partial v}dv) + \frac{\partial z}{\partial y}(\frac{1}{2}du + \frac{1}{2}dv)$

$dz = (\frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{1}{2}\frac{\partial z}{\partial y})du + (\frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{1}{2}\frac{\partial z}{\partial y})dv$

5) проверка 4 и 2

$\frac{\partial z}{\partial x}\frac{\partial x}{\partial y} + \frac{1}{2}\frac{\partial z}{\partial y} = -\frac{1}{2}$

$\frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{1}{2}\frac{\partial z}{\partial y} = \frac{1}{2}$

$\frac{\partial z}{\partial x} = \frac{1}{\frac{\partial x}{\partial v} - \frac{\partial x}{\partial u}}$

$\frac{\partial z}{\partial y} = \frac{\frac{\partial x}{\partial u} + \frac{\partial x}{\partial v}}{\frac{\partial x}{\partial u} - \frac{\partial x}{\partial v}}$

6) 5 б уравнение

$u(\frac{\partial x}{\partial v} - \frac{\partial x}{\partial u})^{-1} + v(\frac{\partial x}{\partial v} - \frac{\partial x}{\partial u})^{-1}(\frac{\partial x}{\partial u} + \frac{\partial x}{\partial v}) = 0$

$v\frac{\partial x}{\partial u} + v\frac{\partial x}{\partial v} = u$

Вопрос
Ответ

№ 51(2)

$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 0$

$x = r\cos\varphi$
 $y = r\sin\varphi$

$u = u(x, y) = \tilde{u}(r, \varphi) = \tilde{u}(r(x, y), \varphi(x, y))$

$\frac{\partial u}{\partial x} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$

$\frac{\partial^2 u}{\partial x^2} = (\frac{1}{r} - \frac{x^2}{r^3})\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial r\partial \varphi}\frac{y}{r^2}) + \frac{2xy}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2}(\frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{y}{r^2})$

$$\frac{\partial u}{\partial x \partial y} = -\frac{x}{r^4} \frac{\partial \tilde{u}}{\partial r} + \frac{x}{r} \left(\frac{\partial^2 \tilde{u}}{\partial r^2} \frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \cdot \frac{x}{r^2} - \left(\frac{1}{r^2} - \frac{2y^2}{r^4} \right) \frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2} \left(\frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{x}{r^2} \right) \right)$$

$$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial \tilde{u}}{\partial \varphi} \frac{\partial \varphi}{\partial y} = \frac{y}{r} \frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2} \frac{\partial \tilde{u}}{\partial \varphi}$$

$$\frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{r} + \frac{y^2}{r^3} \right) \frac{\partial \tilde{u}}{\partial r} + \frac{y}{r} \left(\frac{\partial^2 \tilde{u}}{\partial r^2} \frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \frac{x}{r^2} \right) - \frac{2yx}{r^4} \frac{\partial \tilde{u}}{\partial \varphi} + \frac{x}{r^2} \left(\frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{x}{r^2} \right)$$

$$\begin{aligned} & x^2 \left(\left(\frac{r}{r} - \frac{x^2}{r^3} \right) \frac{\partial \tilde{u}}{\partial r} + \frac{x}{r} \left(\frac{\partial^2 \tilde{u}}{\partial r^2} \frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \frac{y}{r^2} \right) + \frac{2yx}{r^4} \frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2} \left(\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{y}{r} \right) \right) + 2xy \left(-\frac{xy}{r^3} \frac{\partial \tilde{u}}{\partial r} + \right. \\ & \left. + \frac{x}{r} \left(\frac{\partial^2 \tilde{u}}{\partial r^2} \frac{y}{r} + \frac{x}{r^2} \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \right) - \left(\frac{1}{r^2} - \frac{2y^2}{r^4} \right) \frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2} \left(\frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{x}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{y}{r^2} \right) + \right. \\ & \left. + y^2 \left(\left(\frac{1}{r} - \frac{y^2}{r^3} \right) \frac{\partial \tilde{u}}{\partial r} + \frac{y}{r} \left(\frac{\partial^2 \tilde{u}}{\partial r^2} \frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \frac{x}{r^2} \right) - \frac{2yx}{r^4} \frac{\partial \tilde{u}}{\partial \varphi} + \frac{x}{r^2} \left(\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2} \frac{x}{r^2} \right) \right) = 0 \right. \\ & \frac{\partial \tilde{u}}{\partial r} \left(\frac{x^2 y^2}{r^3} - \frac{2x^2 y^2}{r^3} + \frac{x^2 y^2}{r^3} \right) + \frac{\partial \tilde{u}}{\partial \varphi} \left(\frac{2yx^3}{r^4} - \frac{2xy}{r^2} + \frac{4xy^3}{r^4} - \frac{2xy^3}{r^4} \right) + \frac{\partial^2 \tilde{u}}{\partial r^2} \left(\frac{x^4}{r^2} + \frac{2x^2 y^2}{r^2} + \frac{y^4}{r^2} \right) + \\ & + \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi} \left(-\frac{x^3 y}{r^3} - \frac{x^3 y}{r^3} + \frac{2x^3 y}{r^3} - \frac{2xy^3}{r^3} + \frac{xy^3}{r^3} + \frac{xy^3}{r^3} \right) + \frac{\partial^2 \tilde{u}}{\partial \varphi^2} \left(\frac{x^2 y^2}{r^4} - \frac{2x^2 y^2}{r^4} + \frac{x^2 y^2}{r^4} \right) = \\ & = r^2 \frac{\partial^2 \tilde{u}}{\partial r^2} \end{aligned}$$

Knoten: $\nabla f(x^0) = 0$

Darm: $\nabla f(x^0) = 0$

> 0 , $\text{no } x^0 - \min$

< 0 , $\text{no } x^0 - \max$

generally we have $\nabla f(x^0) = 0$

§ 5 N18(2)

$$2x^2 + 2y^2 + u^2 + 8y - u + 8 = 0$$

$$u(x, y) = ?$$

$$u'_x = -\frac{F'_x}{F'_u} = \frac{-4x}{2u + 8y - 1} \quad 2u + 8y \neq 1$$

$$u'_y = -\frac{F'_y}{F'_u} = \frac{-4y + 8}{2u + 8y - 1}$$

$$u_x = 0 \Rightarrow x = 0$$

$$y = -2u$$

$$2 \cdot 4u^2 + u^2 + 8u(-2u) - u + 8 = 0$$

$$-7u^2 - u + 8 = 0$$

$$\begin{cases} u_1 = 1 \\ u_2 = -\frac{8}{7} \end{cases}$$

$$(x, y) = (0; \frac{16}{7}) \quad u = -\frac{8}{7}$$

$$(x, y) = (0; -2) \quad u = 1$$

$$u''_{xx} = - \frac{4(2u+8y-1) - u_x(2u'_x)}{(2u+8y-1)^2}$$

$$u''_{yx} = \frac{u_x(2u'_y+8)}{(2u+8y-1)^2}$$

$$u''_{yy} = \frac{(u+8u'_y)(2u+8y-1) - (u+8u'_y)(2u'_y+8)}{(2u+8y-1)^2}$$

$$u''_{xx}|_{(0;-2)} = \frac{-4(2-16-1) - 0}{(1-16-1)^2} = \frac{60}{16^2}$$

$$u''_{xy}|_{(0;-2)} = 0$$

$$u''_{yy} = \frac{-(4+0)(2-16-1) - (-8+8)(0)}{16^2} = \frac{60}{16^2}$$

$$J^2 u|_{(0;-2)} = \frac{60}{16^2} x^2 + \frac{60}{16^2} y^2 \leftarrow \text{Этот min}$$

26.09

Условный экстремум

1] на области $G \subset \mathbb{R}^n$ заданы $f(x); \varphi_1(x); \dots; \varphi_m(x) \quad m < n$

2] нп-во $E = \{x \in \mathbb{R}^n \mid \varphi_1(x) = \dots = \varphi_m(x) = 0\}$

Опр. Точка $x^0 \in G$ наз. (строгой) условной макс. ф-ции f при условиях $\varphi_1(x) = \dots = \varphi_m(x) = 0$, если $\exists \delta > 0: \forall x \in \dot{U}_\delta(x^0) \cap E \rightarrow f(x) \leq f(x^0)$

Необх. усл. условного экстр.

1] $f(x), \varphi_1(x); \dots; \varphi_m(x)$ нем. глуп в $U(x^0)$ и при этом $\text{rank} \begin{pmatrix} \partial \varphi_i \end{pmatrix} (x^0) = m$

Если x^0 абс. усл. экстр., то $\frac{\partial L}{\partial x_i}(x^0) = 0 \quad i = \overline{1; n}$
 $\varphi_i(x^0) = 0 \quad i = \overline{1; m}$

$$(L = f(x) + \sum_{i=1}^m \lambda_i \varphi_i(x); \lambda_i \in \mathbb{R})$$

Достаточ. усл. экстр.

1] $f; \varphi_1; \dots; \varphi_m$ глупы в $U(x^0)$ и локал. необход. условия. Если при этом

$$d\varphi_i(x^0) = \sum_{j=1}^m \frac{\partial \varphi_i}{\partial x_j}(x^0) dx_j = 0$$

$i = \overline{1, m}$ в.ф. d^2L в.ф.:

- а) "+" оц $\Rightarrow x^0$ - точка для min
 б) "-" оц $\Rightarrow x^0$ - точка для max
 в) не знаем $\Rightarrow x^0$ не для экстр

§5 №21(2)

$$u = 1 - 4x - 8y \quad x^2 - 8y^2 = 8$$

$$L = 1 - 4x - 8y + \lambda(x^2 - 8y^2 - 8)$$

набх для

$$\frac{\partial L}{\partial x} = -4 + 2\lambda x = 0$$

$$\frac{\partial L}{\partial y} = -8 - 16\lambda y = 0$$

$$x^2 - 8y^2 - 8 = 0$$

$$\lambda \neq 0 \quad x = \frac{2}{\lambda} \quad y = -\frac{1}{2\lambda}$$

$$\frac{4}{\lambda^2} - 8 \frac{1}{4\lambda^2} = 8$$

$$\frac{2}{\lambda^2} = 8 \quad \lambda = \pm \frac{1}{2}$$

1) $\lambda = \frac{1}{2} \quad (4; -1)$

2) $\lambda = -\frac{1}{2} \quad (-4; 1)$

КАНДИДАТЫ

ВЫБОРЫ ПРОШЛИ,
КАНДИДАТЫ ОСТАЛИСЯ

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Далее для

$$\frac{\partial^2 L}{\partial x^2} = 2\lambda$$

$$\frac{\partial^2 L}{\partial x \partial y} = 0$$

$$\frac{\partial^2 L}{\partial y^2} = -16\lambda$$

$$d^2L = (dx dy) \begin{pmatrix} 2\lambda & 0 \\ 0 & -16\lambda \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = 2\lambda dx^2 - 16\lambda dy^2 = 2\lambda(dx^2 - 8dy^2) \Leftrightarrow$$

$$d\varphi = 0$$

$$\varphi = x^2 - 8y^2 - 8$$

$$2x dx - 16y dy = 0$$

$$x dx - 8y dy = 0$$

$$dx = 8 \frac{y}{x} dy$$

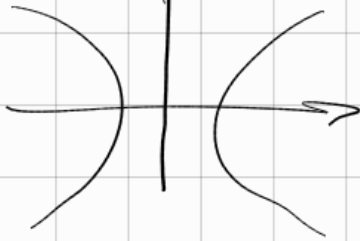
$$\Leftrightarrow 2\lambda(8 \frac{y^2}{x^2} - 8) dy^2$$

$$J^2 L|_1 = 2 \cdot \frac{1}{2} (64 \cdot \frac{1}{16} - 8) dy^2 = -4 dy^2 < 0 \Rightarrow (4, -1) - \text{т. max} \quad u = -7$$

$$J^2 L|_2 = 2 \cdot \frac{1}{2} (64 \cdot \frac{1}{16} - 8) dy^2 = 4 dy^2 > 0 \Rightarrow (1, 1) - \text{т. min} \quad u = 9$$

Второй способ

$$x^2 - 2y = 8 \quad u = 1 - 4x - 2y$$



$$1) \quad x = 2\sqrt{2} \operatorname{ch} t \quad t \in \mathbb{R}$$

$$y = \operatorname{sh} t$$

$$u = 1 - 8\sqrt{2} \operatorname{ch} t - 2 \operatorname{sh} t$$

$$u'_t = 0 \dots \text{Экстремумы по u и y имеют одинаковый знак}$$

$$2) \quad x = -2\sqrt{2} \operatorname{ch} t \quad t \in \mathbb{R}$$

$$y = \operatorname{sh} t$$

$$u = 1 + 8\sqrt{2} \operatorname{ch} t - 2 \operatorname{sh} t$$

$$u'_t = \dots 0$$

§5 126(3)

$$u = xg + yz$$

$$x^2 + y^2 = 0$$

$$y + z = 2$$

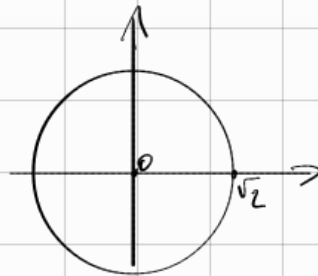
$$y \geq 0$$

$$z = 2 - y$$

$$u = xg + 2y - y^2$$

$$x^2 + y^2 = 2$$

$$y \geq 0$$



$$x = \sqrt{2} \cos \varphi$$

$$y = \sqrt{2} \sin \varphi$$

$$\varphi \in (0; \pi)$$

$$u = 2 \cos \varphi \sin \varphi + 2\sqrt{2} \sin \varphi - 2 \sin^2 \varphi$$

$$u'_\varphi = 2 \cos 2\varphi + 2\sqrt{2} \cos \varphi - 4 \sin \varphi \cos \varphi = 0$$

$$2 \cos^2 \varphi - 1 + \sqrt{2} \cos \varphi - 2 \sin \varphi \cos \varphi = 0$$

$$\sin \varphi = \frac{2 \cos^2 \varphi + \sqrt{2} \cos \varphi - 1}{2 \cos \varphi}$$

$$\frac{4 \cos^4 \varphi + 2 \cos^2 \varphi + 1 - 4 \cos^2 \varphi - 2\sqrt{2} \cos \varphi + 4\sqrt{2} \cos^3 \varphi}{4 \cos^2 \varphi} + \cos^2 \varphi = 1$$

$$8 \cos^4 \varphi - 6 \cos^2 \varphi + 4\sqrt{2} \cos^3 \varphi - 2\sqrt{2} \cos \varphi + 1 = 0$$

$$8t^4 + 4\sqrt{2}t^3 - 6t^2 - 2\sqrt{2}t + 1 = 0$$

$$z = t\sqrt{2} \quad t = z/\sqrt{2}$$

$$2z^4 + 2z^3 - 3z^2 - 2z + 1 = 0$$

$$(z^2 - 1)(2z + 2z - 1) = 0$$

$$\begin{cases} z = \pm 1 \\ z = \frac{-1 \pm \sqrt{3}}{2} \end{cases}$$

Формула Жюлье-Гесса
(Мануэла Гесса)

Формула Дюрера
Решение ???

$$\begin{cases} t = \frac{1}{\sqrt{2}} \\ t = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}} \end{cases}$$

$$\begin{cases} \cos \varphi = \frac{1}{\sqrt{2}} \\ \cos \varphi = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}} \end{cases}$$

$$(1) \begin{cases} x=1 \\ y=1 \end{cases}$$

$$(2) \begin{cases} x=-1 \\ y=1 \end{cases}$$

$$(3) \begin{cases} x = \frac{-1-\sqrt{3}}{2} \\ y = \frac{\sqrt{3}-1}{2} \end{cases}$$

$$(4) \begin{cases} x = \frac{-1+\sqrt{3}}{2} \\ y = \dots \end{cases}$$

$$\begin{aligned} u''_{\varphi\varphi} &= -4\sin\varphi\cos\varphi + 4\cos\varphi(-\sin\varphi) + 2\sqrt{2}(-\sin\varphi) - 4\cos\varphi\cos\varphi - 4\sin\varphi(-\sin\varphi) = \\ &= -4\sin 2\varphi - 2\sqrt{2}\sin\varphi - 4\cos 2\varphi \\ \varphi &= \frac{\pi}{4} \end{aligned}$$

$$u''_{\varphi\varphi} \Big|_{\frac{\pi}{4}} = -4 - 2 - 0 = -6 < 0 \Rightarrow (1;1) \text{ т. max}$$

§5 №31(3)

$$u = x + y + z \quad x^2 + y^2 \leq z \leq 1$$

Найти наиб и наим значения

1) Проверка экстремум функции

$$\nabla u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow \text{нет экстремумов функции}$$

1. "Крайности" $z=1, x^2+y^2 \leq 1$

$$u = x + y + 1 \quad x^2 + y^2 \leq 1$$



проверка функции на экстр

$$\nabla u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

нет экстр

на границе $u = x + y + 1 \quad x^2 + y^2 = 1$

$$x = \cos \varphi \quad y = \sin \varphi$$

$$u = \cos \varphi + \sin \varphi + 1$$

$$u' = -\sin \varphi + \cos \varphi = 0$$

$$\sin \varphi = \cos \varphi \quad \varphi = \frac{\pi}{4} + \pi k$$

$$\begin{cases} x = \frac{1}{\sqrt{2}} \\ y = \frac{1}{\sqrt{2}} \end{cases} \quad u = 1 + \sqrt{2}$$

$$\begin{cases} x = -\frac{1}{\sqrt{2}} \\ y = -\frac{1}{\sqrt{2}} \end{cases} \quad u = 1 - \sqrt{2}$$

2) "Мажоранса"

$$x^2 + y^2 = z \quad u = x + y + z$$

$$x^2 + y^2 \leq 1$$

$$\begin{aligned} x^2 + y^2 &= z \\ x^2 + y^2 &\leq 1 \end{aligned}$$

$$\begin{aligned} x^2 + y^2 &= z \\ x^2 + y^2 &= 1 \end{aligned} \quad \text{мажоранса}$$

$$u = x + y + x^2 + y^2$$

$$x^2 + y^2 \leq 1$$

$$\nabla u = \begin{pmatrix} 1+2x \\ 1+2y \end{pmatrix} = 0 \quad \begin{cases} x = -\frac{1}{2} \\ y = -\frac{1}{2} \end{cases}$$

$$u = -\frac{1}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = -\frac{1}{2}$$

$$\begin{aligned} \eta &\leq 1 + \sqrt{2} \\ \eta &= -\frac{1}{2} \end{aligned}$$